



A constitutive model based on the modified generalized three-dimensional Hoek–Brown strength criterion

Hehua Zhu^{a,b}, Qi Zhang^{c,*}, Boqi Huang^{a,b}, Lianyang Zhang^{d,**}

^a Key Laboratory of Geotechnical and Underground Engineering (Tongji University), Ministry of Education, Shanghai, China

^b Department of Geotechnical Engineering, Tongji University, Shanghai, China

^c School of Civil Engineering, Southeast University, Nanjing, China

^d Department of Civil Engineering and Engineering Mechanics, University of Arizona, Tucson, AZ, USA

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ABSTRACT

A new constitutive model is proposed based on the generalized three-dimensional (3D) Hoek–Brown strength criterion which was proposed by the authors in 2007 and 2008 and later modified in 2013. The model involves a 3D continuous multi-segment plastic flow rule that can not only consider the effect of confining stress on the plastic flow and volumetric deformation, but also avoid the use of uncertain parameters such as dilatancy angle. The constant volume and maximum tensile stress flow rules are adopted at the high compressive confining stress condition, and the tensile stress condition, respectively. A new interpolation between the high compressive confining stress condition and the tensile stress condition is proposed to ensure the continuity of the plastic potential function over the entire stress space. The proposed constitutive model is based on the assumption that the rock mass is elastic-perfectly plastic but can be extended to the case when a rock mass shows strain-hardening behavior. The new constitutive model has been successfully implemented in a 3D finite element code, and validated by analyzing a circular ring under plane-strain loading condition and a model test tunnel with detailed instrumentation and measurements and then comparing the numerical results with the corresponding analytical solutions and experimental results. Finally, the new constitutive model is successfully applied to analyze a real highway tunnel excavated in rock mass with detailed instrumentation and measurements. The obtained tunnel roof and horizontal convergence displacements using the new constitutive model based on the modified GZZ criterion are in good agreement with those from the field measurements, further proving the importance in considering the contribution of intermediate principal stress to rock strength.

1. Introduction

Since the Hoek–Brown (HB) strength criterion was developed for intact rock by Hoek and Brown^{1,2} and later extended to rock masses by Hoek and Brown² and Hoek et al.⁴, it has been widely used in rock engineering due to the adequacy of its predictions of rock behavior and the convenience of its applications to a range of rock engineering problems. However, the HB strength criterion does not consider the effect of the intermediate principal stress although much evidence has been accumulated to indicate that the intermediate principal stress does influence the rock strength in many instances. To overcome this limitation, several three-dimensional (3D) versions of the HB strength criterion have been proposed by researchers.^{5–10} Of these 3D HB strength criteria, only the criterion proposed by Zhang and Zhu⁸ and Zhang⁹ can predict the same strength as the original HB strength

criterion at both triaxial compression and extension states and is considered a true 3D version of the original HB strength criterion. The generalized 3D HB strength criterion proposed by Zhang and Zhu was named GZZ strength criterion by Priest.¹¹ For simplicity, the term “GZZ strength criterion” will be used in this paper.

However, the GZZ strength criterion may have problems with some stress paths and cause inconvenience in numerical applications owing to the non-smoothness of the failure surface at both triaxial compression and extension states and the non-convexity of the failure surface at the triaxial extension state. So Zhang et al.¹² modified the GZZ strength criterion by utilizing three different Lode dependences with characteristics of both smoothness and convexity to replace its Lode dependence. The modified GZZ strength criterion not only maintains the advantages of the GZZ strength criterion, but also solves the non-smoothness and non-convexity problems with no loss of accuracy for

* Corresponding author.

** Corresponding author.

E-mail addresses: zhangqi@seu.edu.cn (Q. Zhang), lyzhang@email.arizona.edu (L. Zhang).

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strength prediction. Therefore, the modified GZZ strength criterion can be used as a failure criterion to construct a new constitutive model for rock masses and implemented in a 3D finite element (FE) code for 3D numerical analysis of rock engineering problems.

Several 3D numerical analysis codes including FLAC3D, PLAXIS and ZSoil are widely used in geotechnical engineering; but these codes only use the original HB strength criterion and thus virtually only 2D rock mass behavior is considered. Also, most of the 3D codes use the HB strength criterion with $a = 0.5$ and thus cannot consider rock masses with $a \neq 0.5$ frequently encountered in practice, where a is a constant reflecting the characteristics of rock masses as described in detail in next section. Therefore, in this paper, a new constitutive model based on the modified GZZ strength criterion is proposed for truly 3D analysis of rock engineering problems. The new constitutive model involves a 3D continuous multi-segment plastic flow rule and has been successfully implemented in the 3D FE code GeoFBA3D. The implemented model is verified by analyzing a circular ring under plane-strain loading condition and a model test tunnel with detailed instrumentation and measurements and then comparing the numerical results with the corresponding analytical solutions and experimental results. Finally, the implemented model is successfully used to analyze a real highway tunnel excavated in rock mass.

2. Modified generalized 3D HB strength criterion

This section briefly describes the modified generalized 3D HB strength criterion (i.e. the modified GZZ strength criterion) in order to provide the required background for the development of the new constitutive model in next section.

For intact rock, the original HB strength criterion^{1,2} may be expressed in the following form

$$\sigma_1 = \sigma_3 + \sigma_c (m_i \frac{\sigma_3}{\sigma_c} + 1)^{0.5} \quad (1)$$

where σ_c is the unconfined compressive strength of the intact rock; σ_1 and σ_3 are respectively the major and minor effective principal stresses; and m_i is a material constant for the intact rock, which depends upon the rock type (texture and mineralogy) and can be obtained from the empirical value table.^{13,14}

For jointed rock masses, the generalized form of the HB criterion,^{3,4} which incorporates both the original and the modified forms, is given by

$$\sigma_1 = \sigma_3 + \sigma_c (m_b \frac{\sigma_3}{\sigma_c} + s)^a \quad (2)$$

where m_b is the material constant for the rock masses; and s and a are constants that depend on the characteristics of the rock masses.

Hoek et al.¹⁵ proposed relationships between m_b , s , a , and Geological Strength Index (GSI) by introducing a new parameter D which is a factor that depends on the degree of disturbance due to blast damage and stress relaxation. The values of D range from 0 for undisturbed in situ rock masses to 1 for very disturbed rock masses.

$$m_b = \exp\left(\frac{\text{GSD} - 100}{28 - 17D}\right) m_i \quad (3a)$$

$$s = \exp\left(\frac{\text{GSD} - 100}{9 - 3D}\right) \quad (3b)$$

$$a = 0.5 + \frac{1}{6}[\exp(-\text{GSI}/15) - \exp(-20/3)] \quad (3c)$$

In order to take account of the effect of the intermediate principal stress, Zhang and Zhu⁸ proposed a 3D version of the HB strength criterion for intact rock and rock mass with $a = 0.5$ by combining the general Mogi¹⁶ criterion and the HB strength criterion. Later, Zhang⁹ extended the 3D HB strength criterion to the general case with any a value, which is expressed as

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \frac{m_b}{2} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right) - m_b \sigma_{m,2} = s \sigma_c \quad (4)$$

where τ_{oct} is the octahedral shear stress and $\sigma_{m,2}$ is the mean stress defined respectively by

$$\tau_{\text{oct}} = \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}, \quad \sigma_{m,2} = \frac{\sigma_1 + \sigma_3}{2} \quad (5)$$

in which σ_1 , σ_2 and σ_3 are, respectively, the major, intermediate and minor effective principal stresses.

The criterion proposed by Zhang and Zhu⁸ and Zhang⁹ was named GZZ strength criterion by Priest¹¹ and recommended as an International Society for Rock Mechanics (ISRM) suggested method in “Three-dimensional failure criteria based on the HB criterion”.¹¹

To address the non-smoothness and non-convexity problems for the GZZ strength criterion, Zhang et al.¹² used three Lode dependences with smoothness and convexity characteristics, the elliptical dependence,¹⁷ the hyperbolic dependence¹⁸ and the spatial mobilized plane dependence,¹⁹ to replace the Lode dependence of the GZZ strength criterion. According to the evaluation of the modified GZZ criteria,^{12,20,21} the criterion modified using the hyperbolic dependence or the spatial mobilized plane dependence has relatively better prediction accuracy than that using the elliptical dependence for both intact rocks and jointed rock masses. Moreover, the criterion modified using the spatial mobilized plane dependence may cause some difficulty in judging the stress state when the Lode angle θ_σ is near 0. Therefore, the modified GZZ criterion with the hyperbolic dependence will be used for the development of the constitutive model in this paper and is presented below.

The modified GZZ strength criteria with the hyperbolic dependence can be expressed as

$$\sqrt{J_2} = L(\theta_\sigma)_H \sqrt{J_{\max}} \quad (6)$$

where J_2 is the second deviatoric stress invariant and defined by $J_2 = [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]/6$; the subscript H stands for the hyperbolic dependence which is expressed as¹⁸

$$L(\theta_\sigma)_H = \frac{2\delta(1 - \delta^2)\cos(\pi/6 - \theta_\sigma) + \delta(\delta - 2)\sqrt{4(\delta^2 - 1)\cos^2(\pi/6 - \theta_\sigma) + (5 - 4\delta)}}{4(1 - \delta^2)\cos^2(\pi/6 - \theta_\sigma) - (\delta - 2)^2} \quad (7)$$

in which, the aspect ratio δ and the Lode angle θ_σ are defined as

$$\delta = \sqrt{J_{\min}/J_{\max}} = \sqrt{J_2(-\pi/6)/J_2(\pi/6)}, \quad \theta_\sigma = \tan^{-1} \left[\frac{\sigma_1 + \sigma_3 - 2\sigma_2}{\sqrt{3}(\sigma_1 - \sigma_3)} \right] \quad (8)$$

where $J_{\max}$ and $J_{\min}$ are the maximum and minimum values of J_2 on the same deviatoric stress plane [i.e. at the same first stress invariant ($I_1 = \sigma_1 + \sigma_2 + \sigma_3$)], respectively. $J_{\max}$ is equal to J_2 when $\theta_\sigma = \pi/6$ at triaxial compression state, and $J_{\min}$ is equal to J_2 when $\theta_\sigma = -\pi/6$ at triaxial extension state.

For a rock mass with $a = 0.5$, the explicit expression of $J_{\max}$ and $J_{\min}$ can be derived as

$$\sqrt{J_{\max}} = \frac{1}{6\sqrt{3}} (\sqrt{m_b^2 \sigma_c^2 + 12m_b \sigma_c I_1 + 36s \sigma_c^2} - m_b \sigma_c) \quad (9)$$

$$\sqrt{J_{\min}} = \frac{1}{3\sqrt{3}} (\sqrt{m_b^2 \sigma_c^2 + 3m_b \sigma_c I_1 + 9s \sigma_c^2} - m_b \sigma_c) \quad (10)$$

Then Eqs. (9) and (10) can be used to calculate the aspect ratio with Eq. (8) and then Eq. (6) can be directly used to determine the strength with given σ_2 and σ_3 . For a rock mass with $a \neq 0.5$, the numerical solution procedure below can be used¹²:

- Assume an initial σ'_1 with known σ_2 and σ_3 , say $\sigma'_1 = \sigma_2$;
- Calculate first stress invariant I_1 using the input data σ_2 and σ_3 ;

- Calculate $J_{\max}$ and $J_{\min}$ with the GZZ strength criterion [Eq. (4)] with known m_b , s , a using $\theta_\sigma = \pi/6$ for $J_{\max}$ and $\theta_\sigma = -\pi/6$ for $J_{\min}$;
- Calculate aspect ratio δ with Eq. (8);
- Calculate σ_1 with Eq. (6) using the Lode dependence from Eq. (7);
- Check the error defined as below

$$|\sigma_1 - \sigma'_1| < \varepsilon \quad (11)$$

when ε is the prescribed convergence limit, say 0.001 MPa;

- If Eq. (11) is satisfied, the obtained σ_1 is the predicted strength at given σ_2 and σ_3 with the modified GZZ criterion. If Eq. (11) is not satisfied, the obtained σ_1 is used as the σ'_1 and the previous steps are repeated until Eq. (11) is satisfied.

3. A new continuous multi-segment plastic flow rule

The new continuous multi-segment plastic flow rule is proposed by extending the work by Carranza-Torres and Fairhurst²² and Cundall et al.²³ which is briefly described below.

Carranza-Torres and Fairhurst²² proposed a linear non-associated plastic flow rule which is a function of mobilized dilatancy angle ψ_{mob} based on the original HB criterion:

$$Q_1 = R_1 - \left(\frac{1 + \sin \psi_{mob}}{1 - \sin \psi_{mob}} \right) R_3 \quad (12)$$

where

$$R_i = \frac{-\sigma_i}{m_b \sigma_c} + \frac{s}{m_b^2} \quad (i = 1, 3) \quad (13)$$

The mobilized dilatancy angle ψ_{mob} varies with the confining stress σ_3 ^{24,25}:

$$\psi_{mob} = \begin{cases} \psi + \frac{\sigma_3}{\sigma_t} \left(\frac{\pi}{2} - \psi \right) & (\sigma_t \leq \sigma_3 < 0) \\ \frac{\sigma_3^{cv} - \sigma_3}{\sigma_3^{cv}} \psi & (0 \leq \sigma_3 \leq \sigma_3^{cv}) \\ 0 & (\sigma_3 \geq \sigma_3^{cv}) \end{cases} \quad (14)$$

where σ_t is the tensile strength equal to $-s\sigma_c/m_b$; σ_3^{cv} is a confining stress beyond which ψ_{mob} is 0 (i.e. no volume change); and ψ is the initial dilatancy angle or the dilatancy angle under uniaxial condition.

Cundall et al.²³ commented that it is not meaningful to use dilatancy angle for rock when the confining stress is low or tensile, because the mode of failure is typically axial splitting instead of shearing. So they proposed a multi-segment plastic flow rule as a function of confining stress without using the dilatancy angle. Instead, a flow factor γ defined as the rate of plastic strain increments in the principal stress directions was introduced to describe the flow rule at different segments. In the vicinity of uniaxial stress condition ($\sigma_3 \approx 0$), the associated flow rule is applied to consider the high rate of volumetric expansion as observed in uniaxial compression tests. The vector of plastic strain rate is normal to the yield surface and thus the highest volumetric strain rate is obtained. Differentiating the expression of the generalized HB strength criterion [Eq. (2)], γ at $\sigma_3 = 0$ is expressed as

$$\gamma_I = -\frac{1}{1 + am_b[m_b(\sigma_3/\sigma_c) + s]^{a-1}} \quad (15)$$

A constant volume flow rule is applied at high compressive confining stress condition, i.e. when the confining stress is beyond the confining stress of constant volume σ_3^{cv} , there is no dilation during the yielding. So when $\sigma_3 \geq \sigma_3^{cv}$, γ is expressed as

$$\gamma_{II} = -1 \quad (16)$$

For $0 \leq \sigma_3 < \sigma_3^{cv}$, γ is linearly interpolated between the associated and constant volume limits and expressed as

$$\gamma_{III} = 1/\left[\frac{1}{\gamma_I} + \left(\frac{1}{\gamma_{II}} - \frac{1}{\gamma_I} \right) \frac{\sigma_3}{\sigma_3^{cv}} \right] \quad (17)$$

When all principal stresses are tensile ($\sigma_t \leq \sigma_3 \leq \sigma_1 < 0$), the flow rate vector is coaxial with the principal stress vector and follows the radial flow rule, and γ is expressed as

$$\gamma_{IV} = \sigma_1/\sigma_3 \quad (18)$$

The plastic flow rule proposed by Cundall et al.²³ can adequately capture the characteristics of rock failure at different confining stress levels. However, the plastic flow rule is discontinuous when it goes from the unconfined ($\sigma_3 = 0$) compressive stress state to the tensile stress zone, and the expressions at the four stress states/zones are not in a unified form. This may lead to inconvenience and singularity in numerical calculations. Also the flow rule by Cundall et al.²³ is for the 2D generalized HB strength criterion. Therefore a new continuous multi-segment plastic flow rule is proposed for both the 2D generalized HB strength criterion and the 3D modified GZZ strength criterion in the following.

3.1. New multi-segment plastic flow rule for generalized HB criterion

Under the condition of associated flow rule, the yield and plastic potential functions for the generalized HB criterion [Eq. (2)] are the same and can be represented as

$$F = Q = \frac{1}{\sigma_c^{1/a-1}} (\sigma_1 - \sigma_3)^{1/a} - m_b \sigma_3 - s\sigma_c \quad (19)$$

Differentiating Eq. (19), the vector of plastic flow is

$$\begin{cases} Q_1 = \frac{\partial Q}{\partial \sigma_1} = \frac{1}{a\sigma_c^{1/a-1}} (\sigma_1 - \sigma_3)^{1/a-1} \\ Q_3 = \frac{\partial Q}{\partial \sigma_3} = -\frac{1}{a\sigma_c^{1/a-1}} (\sigma_1 - \sigma_3)^{1/a-1} - m_b \end{cases} \quad (20)$$

Using flow factor λ , the plastic volumetric strain $d\varepsilon_v^p$ can be expressed as

$$d\varepsilon_v^p = \lambda \left(\frac{\partial Q}{\partial \sigma_1} + \frac{\partial Q}{\partial \sigma_3} \right) = -\lambda m_b \quad (21)$$

So the plastic volumetric strain is only related to m_b which appears just in Q_3 of the vector of plastic flow. By introducing a parameter η in the plastic potential function [Eq. (19)], the linear interpolation between the associated and constant volume flow rules can be obtained, i.e.

$$Q^c = \frac{1}{\sigma_c^{1/a-1}} (\sigma_1 - \sigma_3)^{1/a} - \eta m_b \sigma_3 - s\sigma_c \quad (22)$$

$$\eta = \begin{cases} 1 - \sigma_3/2\sigma_3^{cv} & 0 \leq \sigma_3 \leq \sigma_3^{cv} \\ 0 & \sigma_3 > \sigma_3^{cv} \end{cases} \quad (23)$$

Differentiating Eq. (22), the plastic volumetric strain under compressive condition can be obtained:

$$d\varepsilon_v^p = \lambda \left(\frac{\partial Q^c}{\partial \sigma_1} + \frac{\partial Q^c}{\partial \sigma_3} \right) = \begin{cases} -\lambda (1 - \sigma_3/2\sigma_3^{cv}) m_b & 0 \leq \sigma_3 \leq \sigma_3^{cv} \\ 0 & \sigma_3 > \sigma_3^{cv} \end{cases} \quad (24)$$

When $\sigma_3 = 0$ (uniaxial condition), $d\varepsilon_v^p$ is equal to $-\lambda m_b$ and thus the flow rule is associated, which gives the highest volumetric strain rate. When $\sigma_3 > \sigma_3^{cv}$, $d\varepsilon_v^p$ is equal to 0 and thus the flow rule is constant volume. The coefficient η is called a compressive interpolation parameter.

At tensile stress condition ($\sigma_t \leq \sigma_3 < 0$), the yield and plastic potential functions related to the maximum tensile stress σ_t is adopted, which is similar to the radial flow rule at the maximum tensile condition.^{23,26} The direction of the plastic flow is coaxial with the principal tensile stress vector and the tensile plastic flow Q^t can be expressed as

$$F = Q^t = \sigma_t - \sigma_i \quad (i = 1, 3) \quad (25)$$

Owing to the difference between the expressions in Eqs. (22) and

(25), a direct relationship between the associated and maximum tensile stress flow rules cannot be realized by using the interpolation method. To address this problem, a parameter $\bar{\xi}$ is introduced to modify the tensile plastic flow vector Q^t and consider the effect of the magnitude of σ_1 and σ_3 on the direction of plastic flow:

$$d\epsilon_v^p = \lambda \left(\frac{\partial Q^t}{\partial \sigma_1} + \frac{\partial Q^t}{\partial \sigma_3} \right) = \lambda \left(\bar{\xi}_1 \frac{\partial Q}{\partial \sigma_1} + \bar{\xi}_3 \frac{\partial Q}{\partial \sigma_3} \right) \quad (26)$$

where

$$\bar{\xi}_i = \sqrt{2} \xi_i / (\xi_1^2 + \xi_2^2 + \xi_3^2)^{1/2} \text{ and } \xi_i = \begin{cases} 1 & \sigma_i \geq 0 \\ 1/(1 - \sigma_i/\sigma_t) & \sigma_i < 0 \end{cases} \quad (i = 1, 3) \quad (27)$$

When $\sigma_i < 0$, the coefficient ξ_i and thus the modification of Eq. (26) are activated. Otherwise, ξ_i takes the default value of 1 and no modification is done. Under uniaxial compressive stress condition ($\sigma_1 > 0$, $\sigma_3 = 0$), $\bar{\xi}_1 = \bar{\xi}_3 = 1$ and thus the associated flow rule is not modified. By using $\bar{\xi}$, the continuity between Q^c and Q^t in the vicinity of the uniaxial stress condition is ensured. The coefficient $\bar{\xi}$ can be called a tensile modification parameter. At the maximum tensile stress condition (i.e. when σ_1 and σ_3 are equal to σ_t simultaneously), ξ_i ($i = 1, 3$) are both approaching ∞ and $\bar{\xi}_i$ ($i = 1, 3$) are equal to each other. In this case, the direction of plastic flow is close to that of the resultant of the principal stresses, which is the isocline in the principal stress space.

The new multi-segment plastic flow rule [Eqs. (24) and (26)] can not only consider the effect of confining stress on the plastic flow and volumetric deformation but also avoid the use of uncertain parameters such as dilatancy angle. The two introduced parameters, η and $\bar{\xi}$, can be determined directly using expressions based on the principal stresses. More importantly, the new plastic flow rule ensures the continuity of the plastic potential function in the principal stress space.

3.2. New multi-segment plastic flow rule for modified GZZ criterion

The same procedure as in Section 3.1 is followed to derive the continuous multi-segment plastic flow rule for the modified GZZ criterion. The plastic potential function for the modified GZZ criterion in terms of I_1 , J_2 , and θ_σ can be expressed as:

$$Q = \frac{1}{\sigma_c^{(1/a-1)}} [\sqrt{3} J_2 g(\theta_\sigma)]^{1/a} + \frac{\sqrt{3}}{3} m_b \sqrt{J_2} g(\theta_\sigma) - m_b \frac{I_1}{3} - s\sigma_c \quad (28)$$

where $g(\theta_\sigma) = 1/L(\theta_\sigma)_H$. Under the condition of associated flow rule, the plastic volumetric strain is just related to $\partial Q/\partial I_1$ and can be expressed as:

$$d\epsilon_v^p = \lambda \frac{\partial Q}{\partial I_1} = -\lambda \frac{m_b}{3} \quad (29)$$

So the plastic potential function is controlled by the item $m_b I_1/3$ in Eq. (28). Similarly, by introducing the compressive interpolation parameter η , the plastic potential function Q^c at compressive stress condition can be expressed as:

$$Q^c = \frac{1}{\sigma_c^{(1/a-1)}} [\sqrt{3} J_2 g(\theta_\sigma)]^{1/a} + \frac{\sqrt{3}}{3} m_b \sqrt{J_2} g(\theta_\sigma) - \eta m_b \frac{I_1}{3} - s\sigma_c \quad (30)$$

The η can be determined based on either the first stress invariant I_1 or the minimum principal stress σ_3 . In this paper, the minimum principal stress σ_3 , as in Eq. (23), is directly used for the sake of convenience.

By differentiating Eq. (30), the vector of plastic flow can be obtained:

$$\frac{\partial Q^c}{\partial \sigma_i} = \frac{\partial Q^c}{\partial I_1} \frac{\partial I_1}{\partial \sigma_i} + \frac{\partial Q^c}{\partial \sqrt{J_2}} \frac{\partial \sqrt{J_2}}{\partial \sigma_i} + \frac{\partial Q^c}{\partial J_3} \frac{\partial J_3}{\partial \sigma_i} \quad (31)$$

which can also be expressed in a matrix form as

$$\left[\frac{\partial Q^c}{\partial \sigma_1} \quad \frac{\partial Q^c}{\partial \sigma_2} \quad \frac{\partial Q^c}{\partial \sigma_3} \right] = \left[\frac{\partial Q^c}{\partial I_1} \quad \frac{\partial Q^c}{\partial \sqrt{J_2}} \quad \frac{\partial Q^c}{\partial J_3} \right] [T]_{3 \times 3} \quad (32)$$

where

$$\begin{cases} \frac{\partial Q^c}{\partial I_1} = -\frac{1}{3} \eta m_b \\ \frac{\partial Q^c}{\partial \sqrt{J_2}} = \frac{\sqrt{3} g(\theta_\sigma)}{a\sigma_c^{(1/a-1)}} [\sqrt{3} J_2 g(\theta_\sigma)]^{1/a-1} + \frac{\sqrt{3}}{3} m_b g(\theta_\sigma) \\ \quad - \tan 3\theta_\sigma \left\{ \frac{\sqrt{3}}{a\sigma_c^{(1/a-1)}} [\sqrt{3} J_2 g(\theta_\sigma)]^{1/a-1} + \frac{m_b}{\sqrt{3}} \right\} \frac{\partial g(\theta)}{\partial \theta} \\ \frac{\partial Q^c}{\partial J_3} = -\frac{\sqrt{3}}{2J_2 \cos 3\theta_\sigma} \left\{ \frac{\sqrt{3}}{a\sigma_c^{(1/a-1)}} [\sqrt{3} J_2 g(\theta_\sigma)]^{1/a-1} + \frac{m_b}{\sqrt{3}} \right\} \frac{\partial g(\theta)}{\partial \theta} \end{cases} \quad (33)$$

$$[T]_{3 \times 3} = \begin{bmatrix} 1 & 1 & 1 \\ S_1/2\sqrt{J_2} & S_2/2\sqrt{J_2} & S_3/2\sqrt{J_2} \\ S_2S_3 + J_2/3 & S_1S_3 + J_2/3 & S_1S_2 + J_2/3 \end{bmatrix} \quad (34)$$

in which $S_i = \sigma_i - I_1/3$.

The plastic volumetric strain under compressive stress condition can be obtained as:

$$d\epsilon_v^p = \lambda \left(\frac{\partial Q^c}{\partial \sigma_1} + \frac{\partial Q^c}{\partial \sigma_2} + \frac{\partial Q^c}{\partial \sigma_3} \right) = \begin{cases} -\lambda(1 - \sigma_3/\sigma_3^{cv}) m_b & 0 \leq \sigma_3 \leq \sigma_3^{cv} \\ 0 & \sigma_3 > \sigma_3^{cv} \end{cases} \quad (35)$$

When $\sigma_3 = 0$ (uniaxial condition), $d\epsilon_v^p$ is equal to $-\lambda m_b$ and thus the flow rule is associated. When $\sigma_3 > \sigma_3^{cv}$, $d\epsilon_v^p$ is equal to 0 and thus the flow rule is constant volume.

Similarly, by introducing the tensile modification parameter $\bar{\xi}$, the flow rule Q^t at the tensile stress condition can be obtained:

$$d\epsilon_v^p = \lambda \frac{\partial Q^t}{\partial \sigma_i} = \lambda \bar{\xi}_i \frac{\partial Q}{\partial \sigma_i} \quad (i = 1, 2, 3) \quad (36)$$

where

$$\bar{\xi}_i = \sqrt{3} \xi_i / (\xi_1^2 + \xi_2^2 + \xi_3^2)^{1/2} \text{ and } \xi_i = \begin{cases} 1 & \sigma_i \geq 0 \\ 1/(1 - \sigma_i/\sigma_t) & \sigma_i < 0 \end{cases} \quad (i = 1, 3) \quad (37)$$

When $\sigma_i < 0$, the coefficient ξ_i and thus the modification of Eq. (37) are activated. Otherwise, ξ_i takes the default value of 1 and no modification is done. Under uniaxial compressive stress condition ($\sigma_1 > 0$, $\sigma_2 = \sigma_3 = 0$), $\bar{\xi}_1 = \bar{\xi}_2 = \bar{\xi}_3 = 1$ and thus the associated flow rule is not modified. Using Eq. (37), the effect of tensile stress is considered progressively as can be seen by several typical conditions. At uniaxial tensile stress state ($\sigma_1 = \sigma_2 = 0$, $\sigma_t < \sigma_3 < 0$), ξ_1 and ξ_2 are not activated and thus equal to 1, but ξ_3 is approaching ∞ , leading to $\bar{\xi}_1$ and $\bar{\xi}_2$ both close to 0 and $\bar{\xi}_3$ close to $\sqrt{3}$. So the direction of plastic flow is coaxial with σ_3 . At the maximum tensile stress condition (i.e. when σ_1 , σ_2 and σ_3 are equal to σ_t simultaneously), ξ_i ($i = 1, 2, 3$) are all approaching ∞ and $\bar{\xi}_i$ ($i = 1, 2, 3$) are all equal to 1. In this case, the direction of plastic flow is close to that of the resultant of the principal stresses, which is the isocline in the principal stress space. At other conditions, the direction of the plastic flow is varied according to the magnitude of each principal stress in the tensile state.

The newly developed constitutive models based on the generalized HB criterion and the modified GZZ criterion have been implemented in the 3D FE code GeoFBA3D. To address the issues of singular points in the process of numerical realization, which appear due to the numerical process rather than the continuous multi-segment flow rule, a special numerical treatment is needed. As shown in Eq. (33), when the Lode angle θ_σ is $\pm \pi/6$, the denominator $\cos 3\theta_\sigma$ is equal to 0, leading to the singular problem related to terms $\partial Q/\partial \sqrt{J_2}$ and $\partial Q/\partial J_3$. It is found that the numerator $\partial g(\theta_\sigma)/\partial \theta_\sigma$ is also equal to 0. So the singular points are in the 0/0 indeterminate form and can be addressed using a special numerical method described below.

$\partial g(\theta_\sigma)/\partial \theta_\sigma$ can be obtained as follows:

$$\frac{\partial g(\theta_\sigma)}{\partial \theta_\sigma} = \frac{g_{II}'(\theta_\sigma) g_I(\theta_\sigma) - g_{II}(\theta_\sigma) g_I'(\theta_\sigma)}{g_I^2(\theta_\sigma)} \quad (38)$$

where

$$g_I(\theta_\sigma) = \frac{2\delta(1 - \delta^2)\cos(\pi/6 - \theta_\sigma) + \delta(\delta - 2)}{\sqrt{4(\delta^2 - 1)\cos^2(\pi/6 - \theta_\sigma) + (5 - 4\delta)}} \quad (39)$$

$$g'_I(\theta_\sigma) = \frac{2\delta(1 - \delta^2)\sin(\pi/6 - \theta_\sigma)}{\sqrt{4(\delta^2 - 1)\cos^2(\pi/6 - \theta_\sigma) + (5 - 4\delta)}} + \frac{4\delta(\delta - 2)(\delta^2 - 1)\cos(\pi/6 - \theta_\sigma)\sin(\pi/6 - \theta_\sigma)}{\sqrt{4(\delta^2 - 1)\cos^2(\pi/6 - \theta_\sigma) + (5 - 4\delta)}} \quad (40)$$

$$g_{II}(\theta_\sigma) = 4(1 - \delta^2)\cos^2(\pi/6 - \theta_\sigma) - (\delta - 2)^2 \quad (41)$$

$$g'_{II}(\theta_\sigma) = 8(1 - \delta^2)\cos(\pi/6 - \theta_\sigma)\sin(\pi/6 - \theta_\sigma) \quad (42)$$

With Eqs. (38)–(42) and using the L' Hôpital's rule,²⁷ $[dg(\theta_\sigma)/d\theta_\sigma]/\cos 3\theta_\sigma$ at $\theta_\sigma = \pm \pi/6$ can be determined as

$$\lim_{\theta_\sigma \rightarrow \pi/6} \frac{dg(\theta_\sigma)/d\theta_\sigma}{\cos 3\theta_\sigma} = -\frac{2(1 - \delta^2)}{3\delta(2\delta - 1)} \quad (43)$$

$$\lim_{\theta_\sigma \rightarrow -\pi/6} \frac{dg(\theta_\sigma)/d\theta_\sigma}{\cos 3\theta_\sigma} = \frac{(1 - \delta^2)}{\delta(\delta - 2)^2} \quad (44)$$

The above formulation is based on the assumption that the rock mass is elastic-perfectly plastic. However, in many cases, rock masses may show strain-hardening behavior,^{28–30} especially at high confining stress conditions. The proposed constitutive model can be extended to consider the strain-hardening behavior by introducing a hardening parameter.^{31–34} This will be further pursued in the future.

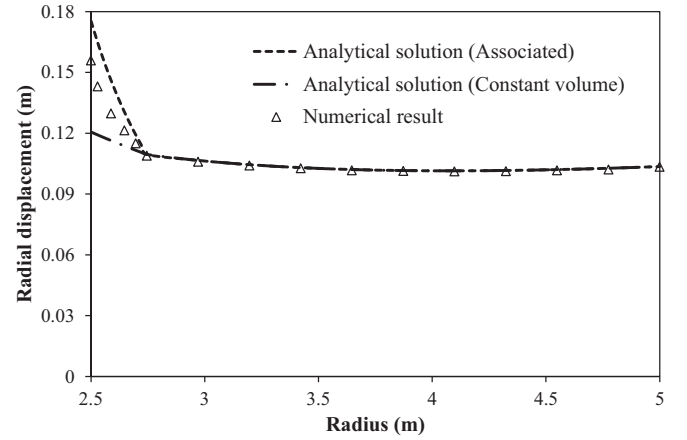
4. Validation of constitutive model

To validate the proposed constitutive model, first a classical mechanics problem, a circular ring under plane-strain loading condition, is analyzed and the results are compared with the derived analytical solutions. Then a model test tunnel with detailed instrumentation and measurements is simulated and the results are compared with the experimental results.

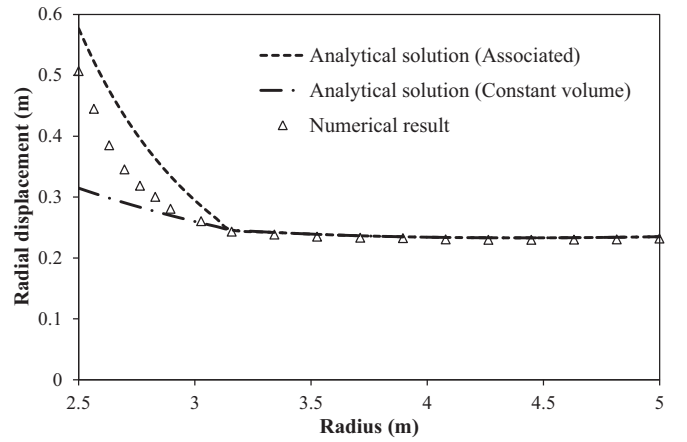
4.1. Circular ring under plane-strain condition

A circular ring under plane-strain condition is a classical problem in elastic-plastic mechanics. It is essentially a simple mechanics model for circular tunnel excavation. The numerical model of the circular ring is produced using GeoFBA3D. The ring has an outer and inner radius of 5 m and 2.5 m, respectively, and a thickness of 0.5 m, and is divided into 177,988 tetrahedron elements. The meshes near the inner circumference are refined to improve the numerical accuracy. The top and bottom surfaces of the circular ring are fixed in the normal direction. The inner circumference of the circular ring is free and the outer circumference of the circular ring is subjected to a confining pressure. The rock mass properties for the circular ring are listed in Table 1.

Numerical analyses are carried out using both the constitutive models based on the original generalized HB criterion and the modified GZZ criterion. Since the analytical solutions based on the modified GZZ criterion cannot be derived, only the numerical results using the constitutive model based on the original generalized HB criterion are compared with the corresponding analytical solutions which are derived following.^{35,36} Fig. 1 compares the radial displacements from the numerical analyses and the analytical solutions at two different confining pressures, 100 MPa and 200 MPa, based on the original



(a) Confining pressure = 100 MPa



(b) Confining pressure = 200 MPa

Fig. 1. Comparison of radial displacements from numerical simulations with those from analytical solutions (using both associated and constant volume flow rules) based on the original generalized HB criterion. The inward displacement is positive.

generalized HB criterion. The analytical solutions are obtained using the associated flow rule and the constant volume flow rule, respectively. In the elastic zone, the numerical results are in close agreement with the analytical solutions. In the plastic zone, the numerical results are smaller than the analytical solutions based on the associated flow rule, but larger than those based on the constant volume flow rule. This is because the numerical results are obtained using the new multi-segment plastic flow rule which is based on both the associated flow rule and the constant volume flow rule.

Since the analytical solutions with the modified GZZ criterion cannot be derived, the numerical results using the constitutive model based on the modified GZZ criterion are only compared with those using the constitutive model based on the original generalized HB criterion. The overall trends of the numerical results from the two constitutive models are very similar to each other. However, the radius of the plastic zone obtained using the constitutive model based on the modified GZZ criterion is smaller than (about half) that obtained using

Table 1

Material properties used in the analyses of a circular ring under plane-strain condition and a physical model tunnel (similarity coefficient = 30).

Property	ρ (kg/m ³)	E (GPa)	ν	m_i	GSI	m_b	s	a	σ_c (MPa)	σ_3^{cr} (MPa)
Circular ring	2300	5.7	0.3	30	100	30	1	0.5	100	100
Physical model tunnel	2100	0.3	0.32	12	30	0.985	4.2×10^{-4}	0.522	0.95	0.95

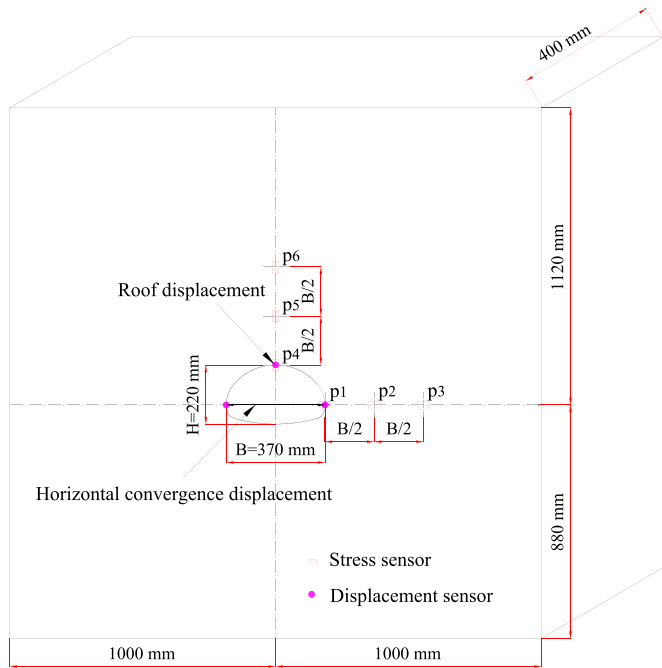


Fig. 2. Sketch of model tunnel test and location of measurement points.

the constitutive model based on the original generalized HB criterion. This is expected because the modified GZZ criterion considers the contribution of the intermediate principal stress to strength.

4.2. Model tunnel test

Vu^{37,38} conducted a series of tests on model tunnels. Fig. 2 shows the sketch of a model tunnel and the location of measurement points. The outer size of the physical model is 2 m × 2 m × 0.4 m and the tunnel size is 370 mm × 220 mm. With a geometric similarity coefficient of 30, the model can simulate a field tunnel at a size of 11.1 m × 6.6 m. Six sensors were installed to measure the stresses, with sensors p1, p2 and p3 measuring the horizontal stresses and p4, p5 and

p6 the vertical stresses. Displacement sensors were also installed to measure the vertical displacement at the roof and the horizontal convergence displacement of the side walls.

The front and back were restricted from normal movement in order to simulate the field 3D plane-strain condition. Loading and unloading at the top and two sides were controlled by individual hydraulic jack systems. By adjusting the applied vertical and horizontal stresses, different lateral coefficients and buried depths can be simulated.

The model tunnel test at a lateral coefficient equal to 1.0 is selected for the validation of the proposed constitutive model. The numerical analysis simulates the five steps used in the physical model test for investigating the effect of buried depth on the tunnel behavior (Table 2):

- Apply an initial pressure on the top and both sides corresponding to a buried depth of 100 m in the field (step 1),
- Excavate the tunnel (step 2),
- Increase the pressure to that corresponding to a buried depth of 400 m in the field (step 3),
- Increase the pressure to that corresponding to a buried depth of 750 m in the field (step 4), and
- Increase the pressure to that corresponding to a buried depth of 1000 m in the field (step 5).

It is noted that applied pressure in the physical model test is 1/30 of the corresponding field pressure due to a geometric similarity coefficient of 30 used. The numerical model which consists of 89,436 tetrahedral elements is produced using GeoFBA3D. The front and back boundaries are fixed at all nodes in the normal directions. The bottom boundary is restricted for vertical movement. The top and two sides are loaded and maintained with the same pressure to simulate a lateral coefficient of 1.0. The same material parameters as in the physical model test (Table 1) are used in the numerical analysis.

As a first comparison, the real-time images obtained in the physical model test by a digital camera and the snapshots from the numerical analysis at the end of steps 2–5 are shown in Table 2. The experimental and numerical results are in good agreement regarding the development of plastic zones. Figs. 3 and 4 further compare the measured stresses and displacements from the physical model test with those

Table 2
Sequence of model tunnel test and behavior of surrounding rock in each step.

Sequence of model tunnel experiment					
	Step 1	Step 2	Step 3	Step 4	Step 5
Field depth	100 m	100 m (Excavation)	400 m	750 m	1000 m
Field stress	2.1 MPa	2.1 MPa (Excavation)	8.4 MPa	15.75 MPa	21 MPa
Applied pressure	70 kPa	70 kPa (Excavation)	280 kPa	525 kPa	700 kPa
Behavior of surrounding rock					
Physical model test (real-time images)					
Numerical analysis (plastic zones)					

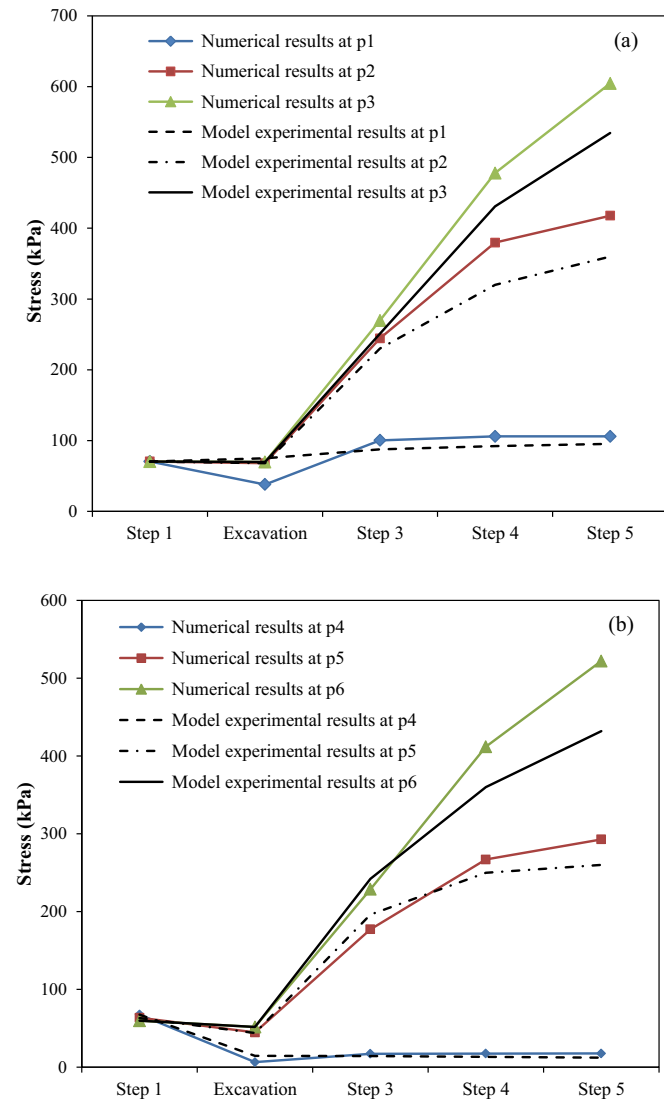


Fig. 3. Comparison of (a) horizontal stresses, and (b) vertical stresses from physical model tunnel test and numerical analysis using the constitutive model based on the modified GZZ criterion.

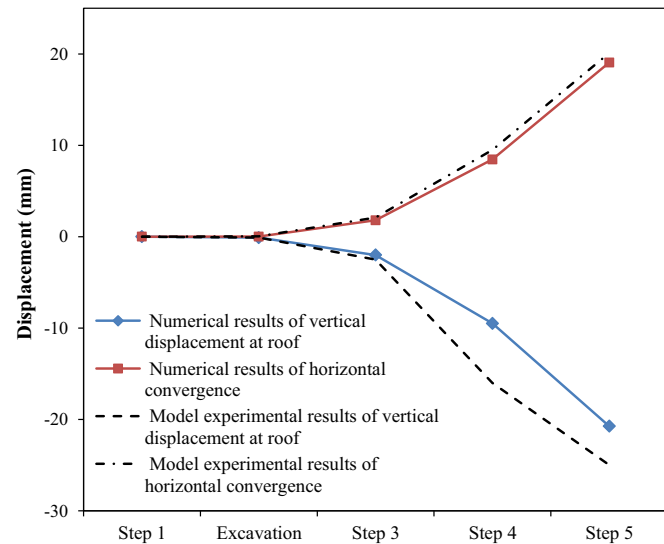
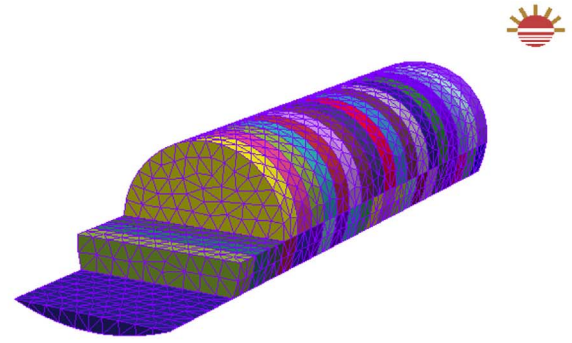
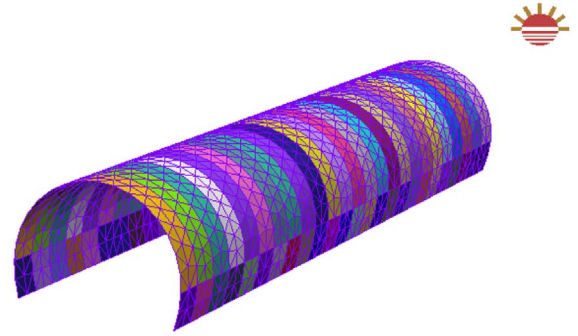


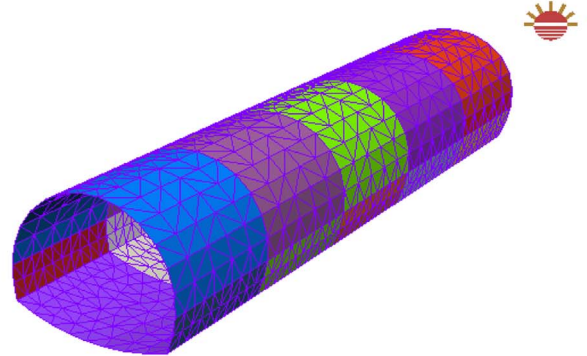
Fig. 4. Comparison of horizontal and vertical displacements from physical model tunnel test and numerical analysis using the constitutive model based on the modified GZZ criterion.



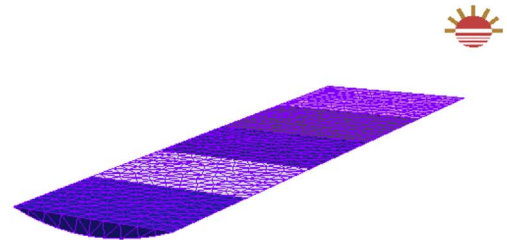
(a) Excavation with bench tunneling construction method



(b) Numerical model of initial lining



(c) Numerical modeling of secondary lining



(d) Numerical modeling of floor backfilling

Fig. 5. Numerical simulation of tunnel construction process.

from the numerical analysis, respectively. There is a very good agreement between the experimental and numerical results until step 3. The larger differences between the experimental and numerical

Table 3
Material properties of initial lining, secondary lining and floor backfilling.

	Material	Density (kg/m ³)	E (MPa)	ν	Thickness (cm)
Initial lining	Plain concrete	2.2×10^3	2.1×10^4	0.25	20
Secondary lining	Reinforced concrete	2.5×10^3	2.95×10^4	0.25	40
Floor backfilling	Plain concrete	2.4×10^3	2.6×10^4	0.25	120 ^a

^a 120 cm is the maximum thickness of the floor backfilling.

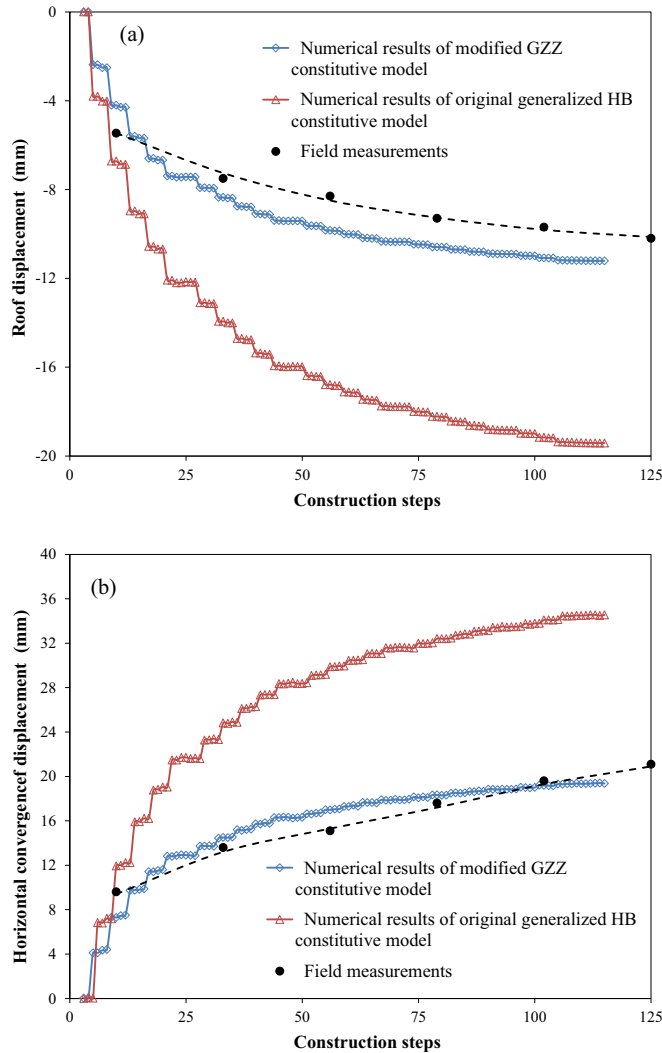


Fig. 6. Comparison of (a) tunnel roof displacements, and (b) tunnel horizontal convergence displacements from numerical analyses and field measurements.

results in steps 4 and 5 are simply due to the fact that the fall of rock at the roof and two shoulders in the physical model test is not considered in the numerical analysis. However, this is simply due to the intrinsic limitation of the FEM rather than the new proposed constitutive model.

5. Application to a highway tunnel excavated in rock mass

An 8 km highway tunnel was constructed in Guizhou province in west China. A 50 m long section of the tunnel at a buried depth of about 140 m is selected for the numerical analysis. The numerical model is $84 \text{ m} \times 60 \text{ m} \times 50 \text{ m}$ and divided into 227,044 tetrahedral elements using GeoFBA3D. The numerical analysis considers 125 steps of construction process, including the up and down bench cut, the initial

lining installation, the secondary lining installation, and the floor backfilling, as shown in Fig. 5. Each bench cut and initial lining is 2 m long, and each secondary lining and floor backfilling is 10 m long. The material properties of the initial lining, secondary lining and floor backfilling are listed in Table 3. In order to simulate the overburden pressure, a 2.2 MPa uniform stress is applied at the top surface of the numerical model. The bottom boundary is fixed at all nodes in the normal direction and the four vertical boundaries are restricted for lateral movement.

The rock at the site is sandstone with unconfined compressive strength σ_c equal to 13.2 MPa. According to Hoek and Brown¹³ and Marinos and Hoek¹⁴ the material constant m_i of 17 is selected. The geological strength index (GSI) of the rock mass at the site is estimated using the method proposed by Sonmez and Ulusay.^{39,40} Based on the distribution of main discontinuities observed on a tunnel face, the rock mass is considered between blocky and very blocky with a structural rating (SR) of 60. The infilling material within the discontinuities is soft clay and less than 5 mm thick. The discontinuities are slightly rough and moderately weathered. So the surface condition rating (SCR) of the discontinuities is about 8. According to the modified quantitative GSI system,⁴⁰ the GSI of the rock mass is about 47. The tunnel was excavated using the blast drilling method which can result in some disturbance to the surrounding rock. So the disturbance factor D is selected to be 0.3. With the obtained m_i , GSI and D , m_b , s and a are determined by using Eq. (3) as 1.834, 0.0014 and 0.507, respectively. The confining stress of constant volume σ_3^{cv} is set to be equal to σ_c . The rock mass is considered linearly elastic-perfectly plastic. The deformation modulus of the rock mass is determined from the GSI using the empirical relation by Hoek and Diederichs⁴¹ as

$$E_m = 100 \left(\frac{1-D/2}{1 + e^{(75+25D-GSI)/11}} \right) = 3.24 \text{ GPa}$$

Fig. 6(a) and (b) show the obtained tunnel roof displacements and horizontal convergence displacements in the 0–2 m section at different construction steps using the constitutive models based on the modified GZZ criterion and the original generalized HB criterion, respectively. For comparison, the displacements from field measurements are also shown in the figure. The tunnel excavation advances about 10 m a day, which corresponds to 23 simulation steps in the numerical analysis. Since the field measurements started after the initial lining of the 6–8 m excavation section was installed (corresponding to simulation step 10) and continued daily (corresponding to every 23 simulation steps), the displacements shown in Fig. 6 are those at simulation steps 10, 33, 56, 79, 102 and 125. The displacements from the numerical analyses using the constitutive model based on the modified GZZ criterion are in good agreement with those from the field measurements not only in general trend but also in the magnitude. The displacements from the numerical analyses using the constitutive model based on the original generalized HB criterion, however, are much larger than those from the field measurements, with the difference getting bigger as construction advances.

Fig. 7 compares the plastic zones from the numerical analyses using the constitutive models based on the original generalized HB criterion and the modified GZZ criterion, respectively. When the original generalized HB criterion is used [Fig. 7(a)], yielding occurs at both the two side walls and the floor of the tunnel. The average radius of the plastic zone at the two side walls is about 4.2 m. When the modified GZZ criterion is used [Fig. 7(b)], however, yielding occurs mainly at the two side walls of the tunnel. The average radius of the plastic zone at the two side walls is merely about 2.0 m and there is no obvious plastic zone at the floor. Again, this is because the modified GZZ criterion considers the contribution of the intermediate principal stress to strength.

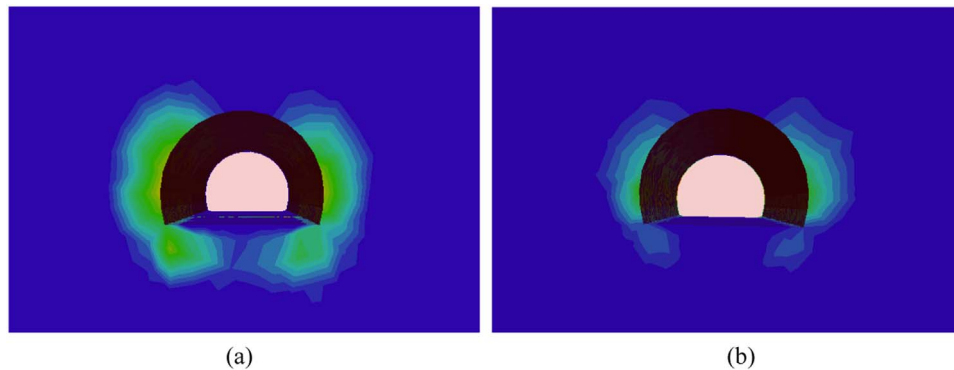


Fig. 7. Comparison of plastic zones of the tunnel from numerical simulations using the constitutive models based on (a) The original generalized HB criterion; and (b) The modified GZZ criterion.

6. Summary and conclusions

The proposed continuous plastic flow rule can not only consider the effect of confining stress on the plastic flow and volumetric deformation but also avoid the use of uncertain parameters such as the dilatancy angle. The two introduced parameters can be determined by directly using expressions related to the state of the confining stress. Further, the new plastic flow rule can ensure the continuity of the plastic potential function at the entire stress space.

The new constitutive model is successfully implemented in 3D FE software GeoFBA3D and validated by analyzing a circular ring under plane-strain condition and a physical model test tunnel with detailed instrumentation and measurements and comparing the numerical results with the corresponding analytical solutions and experimental results.

A real highway tunnel excavated in rock mass is analyzed using the proposed constitutive model. The obtained tunnel roof and horizontal convergence displacements using the new constitutive model based on the modified GZZ criterion are in good agreement with those from the field measurements. The results further show that the displacements and plastic zones from the analysis using the constitutive model based on the modified GZZ criterion are both smaller than those using the constitutive model based on the original generalized HB criterion, further proving the importance in considering the contribution of intermediate principal stress to rock strength.

Acknowledgement

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